# UNIT V: Electrical Systems & Advances

Syllabus: Battery, general electrical circuits, Active Suspension System (ASS), Electronic Brake Distribution (EBD), Electronic Stability Program (ESP), Traction Control System (TCS), GPS, hybrid vehicle, fuel cell.

## 1. Automobile Battery

- The battery is an electrochemical device that stores electrical energy as chemical energy and supplies it to the starter motor, ignition, lights, ECU & accessories when the engine is off or when demand exceeds alternator output.

### Functions

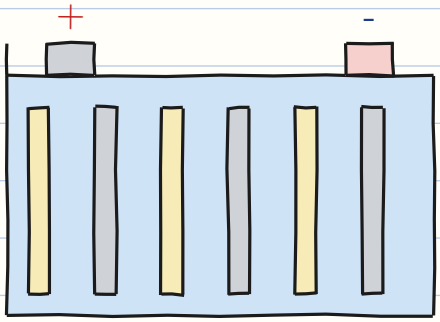
- Crank the engine (starter 100-300 A); supply power with engine off; stabilise voltage (acts as a capacitor for the electrical system); supply extra current at idle.

### Types (automotive)

- Lead-acid (SLI: Starting-Lighting-Ignition) - flooded, VRLA (AGM / gel), maintenance-free; Li-ion / LFP (EV, start-stop, bikes); Ni-MH (hybrids); Ni-Cd (old).

### Lead-acid battery construction

- Container (polypropylene) with 6 cells x 2.1 V = 12.6 V; each cell = positive plates ($PbO_2$, brown) + negative plates (spongy Pb, grey) interleaved with separators (microporous PE / glass mat); electrolyte = dilute sulphuric acid ($H_2SO_4$, sp. gr. 1.28).




+ve plate: lead dioxide
-ve plate: spongy lead
separators (insulator)
electrolyte: dil. $H_2SO_4$
cell connectors, vent plugs

Fig 5.1 Lead-acid battery (section)

### Rating

Capacity in Ah (e.g. 45 Ah @ 20-h rate); CCA (cold cranking amps at -18 deg C); reserve capacity (RC) in minutes.

# 2. Battery - Chemistry & Maintenance

## Chemical reactions

- Discharge: $PbO_2 + Pb + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$ (both plates turn to lead sulphate; acid becomes weaker, specific gravity falls).
- Charge: $2PbSO_4 + 2H_2O \rightarrow PbO_2 + Pb + 2H_2SO_4$ (reverse; acid strength rises; gassing near full charge: $H_2 + O_2$).

SG vs state of charge

| 1.280 | 1.220 | 1.150 |
|---|---|---|
| Fully charged | ~ 50 % charged | Discharged |

measured with hydrometer; temperature correction 0.007 per 10 deg C

Fig 5.2 Specific gravity indicates charge

## Charging

- Alternator charges at 13.8 - 14.4 V (regulated). External charging: constant-current (C/10 = 4.5 A for 45 Ah) or constant-voltage; fast charging only for short time. Avoid sparks - hydrogen gas is explosive.

## Maintenance

- Keep electrolyte level above plates with distilled water (never acid); clean & grease terminals (sulphation); secure battery; check SG monthly; don't leave discharged (sulphation).
- Sulphation: hard $PbSO_4$ crystals on plates (long discharge) -> low capacity. Self-discharge 3-4 % per month. Life 3-5 years.
- Maintenance-free batteries use Ca-alloy grids (low gassing), sealed with magic-eye hydrometer. Connect +ve first, -ve last (remove opposite). Series connection increases voltage, parallel increases capacity.

# 3. Other Batteries (Li-ion, Ni-MH, VRLA)

| Battery | Cell voltage | Energy (Wh/kg) | Features / use |
|---|---|---|---|
| Lead-acid flooded | 2.1 V | 30 - 40 | Cheap, high surge, heavy; SLI |
| VRLA (AGM / gel) | 2.1 V | 30 - 45 | Spill-proof, start-stop vehicles, UPS |
| Ni-Cd | 1.2 V | 40 - 60 | Durable, toxic Cd; old tools |
| Ni-MH | 1.2 V | 60 - 100 | Hybrid cars (Prius), safe, memory effect low |
| Li-ion (NMC / NCA) | 3.6 - 3.7 V | 150 - 250 | EV, high energy, need BMS |
| LFP (LiFePO4) | 3.2 V | 90 - 160 | Long life (3000 cycles), safe; buses, 2W |

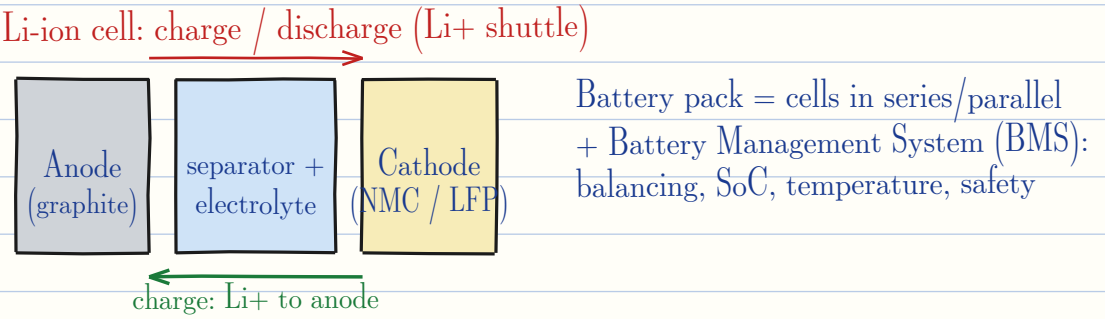



Fig 5.3 Lithium-ion cell principle

## Comparison of lead-acid vs Li-ion

- Li-ion: 3-5x energy density, longer cycle life, faster charging, low self-discharge, but costly, thermal runaway risk, needs BMS.
- Lead-acid: low cost, recyclable (> 95 %), robust, high cold-cranking current; heavy, short life.
- Ultracapacitors (EDLC) - high power, quick charge / discharge; used with batteries for regenerative braking & start-stop.

### 48 V mild hybrid

Uses a 48 V Li-ion battery along with 12 V lead-acid for e-boost, regeneration & stop-start - cheaper than full hybrid.

# 4. General Electrical Circuits

- Vehicle electrical system is a 12 V (24 V for trucks) DC, negative earth (chassis return) system. Main groups: charging, starting, ignition, lighting, instrumentation, accessories, body electronics (BCM), ECU network (CAN).

## Circuit elements

- Source (battery / alternator), conductor (copper wire / wiring harness, colour coded, gauge by current), switch, load (lamp, motor), protection (fuse, fusible link, circuit breaker), relay (low-current switch controlling heavy load), earth return.

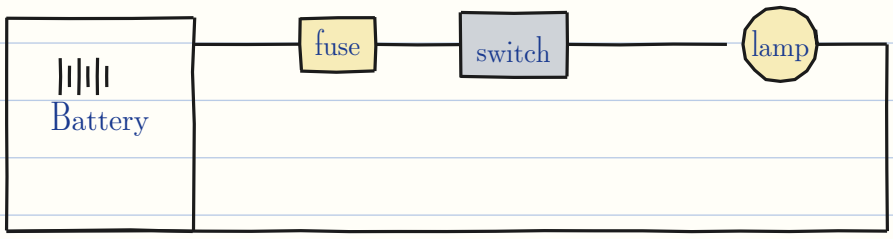



Fig 5.4 Simple vehicle circuit

## Relay circuit

- A small switch energises the relay coil (85-86); the armature closes contacts 30-87 to feed the load (headlamp, horn, fuel pump) from the battery directly -> thinner switch wiring, less voltage drop.

## Basic laws

- Ohm: $V = I\,R$. Power: $P = V\,I$. Fuse rating ~ 125 % of the load current; wire gauge (mm^2) chosen for voltage drop < 0.5 V.

| Fuse colour (blade) | Rating |
|---|---|
| Tan / brown / red | 5 / 7.5 / 10 A |
| Blue / yellow | 15 / 20 A |
| Clear / green | 25 / 30 A |

# 5. Charging, Starting & Lighting Circuits

## Charging system

- Alternator (3-phase AC with rectifier diodes) driven by belt; voltage regulator keeps 13.8 - 14.4 V; charge indicator lamp. Output 60-150 A; field current through slip rings & brushes.

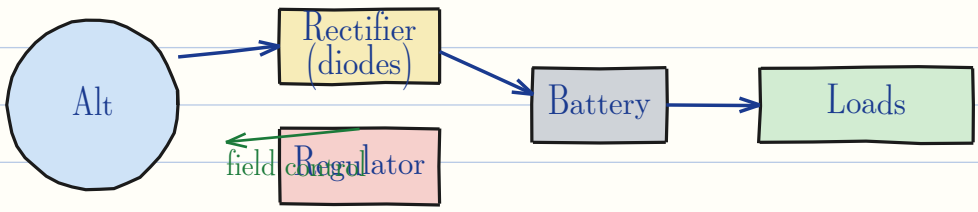



Fig 5.5 Charging circuit

## Starting system

- Battery -> ignition switch -> starter solenoid (relay) -> starter motor (series-wound DC, 1-2 kW) -> pinion engages the flywheel ring gear (ratio ~ 15:1) via Bendix / pre-engaged drive; overrunning clutch protects armature.
- Cranking speed 100-200 rpm; cable thick (low resistance); neutral-safety switch for AT.

## Lighting & auxiliary circuits

- Head lamps (halogen, LED, HID/xenon) high/low beam with dip switch & relay; side / tail lamps; stop lamps (brake switch); indicators with flasher unit (60-120 flashes/min); fog lamps; reverse lamp.
- Auxiliaries: horn, wipers, power windows, fuel gauge, temperature gauge, instrument cluster, central locking, CAN bus networking ECUs (up to 100 per vehicle).

### Starter circuit values

Cranking current 150-300 A (petrol), up to 600 A (diesel); voltage must stay above ~ 9.6 V while cranking. Cable resistance must be below ~ 0.002 ohm.

# 6. Active Suspension System (ASS)

- A passive suspension has fixed spring & damper values - a compromise between comfort and handling. An active suspension uses sensors, ECU and actuators to change forces in real time -> keeps the body level and the tyres in contact on any road.

## Components

- Sensors: wheel / body accelerometers, ride-height (displacement), steering angle, vehicle speed, brake pressure, yaw rate, camera road scan.
- ECU (control algorithm - sky-hook / PID / fuzzy); actuators: hydraulic / electromagnetic / pneumatic at each wheel in series with (or replacing) the spring; pump & accumulator; servo valves.

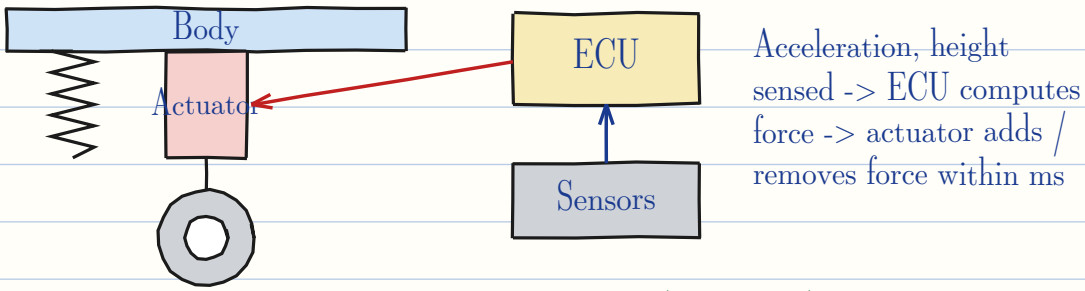



Fig 5.6 Active suspension (one wheel)

## Working

- When the car hits a bump, sensors detect body acceleration; ECU commands the actuator to extend / retract and absorb it. In cornering, braking & acceleration it counteracts roll, dive and squat (body stays flat).

## Advantages / disadvantages

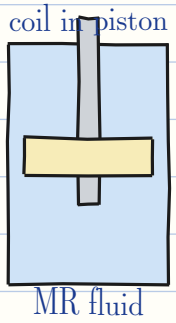
- + Excellent ride, no roll / pitch, self-levelling, adjustable height & firmness, better traction. - Costly, complex, power consumption (~ 5 kW hydraulic), maintenance.
- Examples: Mercedes Active Body Control (ABC) / Magic Body Control, Lotus active, Citroen Hydractive, Bose electromagnetic.

# 7. Semi-Active Suspension & MR Dampers

- Semi-active suspension: springs are passive but the damping force is varied electronically in milliseconds (no big power needs) - cheaper than fully active.

## Types of control

- Electro-hydraulic variable dampers (solenoid valve changes orifice area) - VW DCC, BMW EDC, Bosch CDC.
- Magnetorheological (MR) dampers: fluid with iron particles thickens in a magnetic field -> damping changes in ~ 1 ms (Audi Magnetic Ride, Ferrari, Cadillac MagneRide).




No current: particles free -> fluid thin -> soft damping

Current ON: particles align into chains -> thick fluid -> firm damping

Fig 5.7 Magnetorheological damper

## Air + adaptive systems

- Air spring + CDC (Audi adaptive air suspension; Range Rover): ride height changes (access, off-road, high-speed lowering) plus adaptive damping.
- Predictive systems: camera scans road ahead (Magic Body Control) and prepares the suspension before the bump.

| Type | Spring | Damper | Power use |
|---|---|---|---|
| Passive | Fixed | Fixed | None |
| Semi-active | Fixed | Variable | Low |
| Fully active | Actuator-controlled | Actuator-controlled | High |

# 8. Electronic Brake Distribution (EBD)

- EBD (EBFD) is a software function of the ABS system that automatically varies braking force between front and rear wheels (and left/right) depending on load, deceleration and road grip - replaces the mechanical proportioning valve.

## Why needed?

- During braking, weight shifts forward -> rear wheels become lightly loaded and can lock first (spin-out risk). With a load in the boot, rear axle can take more brake force. A fixed valve cannot adapt to all these conditions.

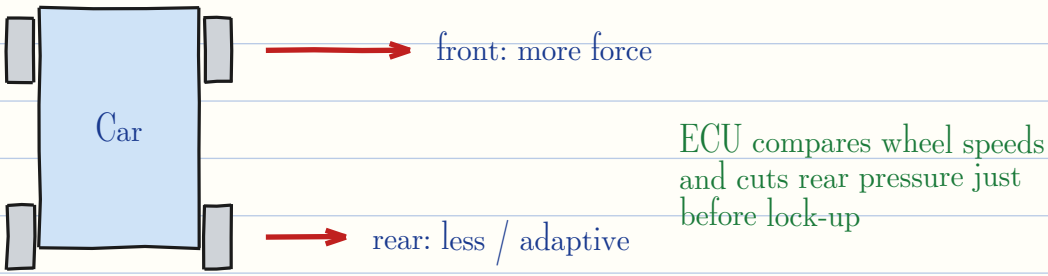



Fig 5.8 Brake force distribution under braking

## Working

1. Wheel speed sensors (ABS) tell the ECU the slip of each wheel.
2. If rear wheel slip exceeds the front by a threshold, ECU closes the rear inlet valve (hold) or releases pressure (like ABS but earlier).
3. Rear brake pressure is limited, front pressure still builds to full -> shortest stop, stable car.

- Advantages: optimum use of rear grip, shorter braking distance, less pedal effort, no valve wear, adapts to load & slope, reduces brake fade at rear. Works before ABS cuts in.

# 9. EBD vs Proportioning Valve & Brake Assist

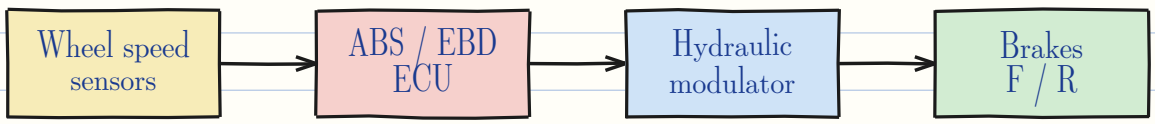
| Point | Mechanical valve | EBD |
|---|---|---|
| Control | Fixed / load-sensing spring | Electronic, real-time |
| Adaptation | Little | Load, road, cornering |
| Cost | Low | Software on ABS hardware |
| Fault display | None | Warning lamp (brake / ABS) |
| Accuracy | Approx. | Precise |

## Cornering Brake Control (CBC)

- EBD extension - during braking in a bend, inside wheel carries less load; the ECU reduces pressure at the inner rear wheel to avoid oversteer / yaw.

## Brake Assist (BA / EBA)

- System detects emergency (panic) braking from the speed at which the pedal is pressed and applies full brake pressure immediately even if the driver does not press hard enough (average drivers use only ~ 40-50 % force). ABS then prevents lock-up.




EBD uses the same hardware as ABS - only extra software

Fig 5.9 Hardware shared by ABS, EBD and BA

## Typical ABS-family hierarchy

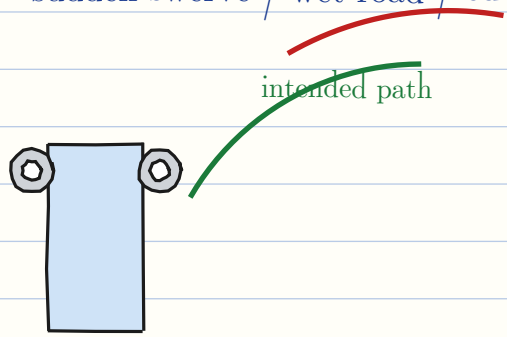
- Conventional brake -> ABS (anti-lock) -> EBD -> BA -> TCS (traction) -> ESP (stability, built on all above). A modern car has them integrated in one hydraulic control unit (HCU).
- Mandatory: ABS in all new cars in India from 2019 (also airbags), EBD with ABS.

# 10. Electronic Stability Program (ESP)

- ESP / ESC / DSC / VSC is an active safety system that detects and reduces loss of traction (skidding) by braking individual wheels and cutting engine torque to keep the car on the driver's intended path.

## Why needed?

- Understeer - front wheels slide, car goes wider than steered (plough). Oversteer - rear slides, car spins inward. Both when the lateral grip limit is exceeded in sudden swerve / wet road / curves.




ESP brakes INNER rear wheel
in understeer -> pulls nose in

ESP brakes OUTER front wheel
in oversteer -> corrects spin

Fig 5.10 ESP corrective action

## Sensors

- Steering angle (driver intent), yaw-rate (rotation about vertical axis), lateral accelerometer, wheel speed (4), brake pressure, throttle / engine torque (CAN).

## Actuators

- Hydraulic unit (ABS valves + active pre-charge pump) - brakes any wheel independently, ECU requests engine torque reduction (throttle / ignition / fuel cut), warning lamp (flashes when active).

## ESP = ABS + TCS + yaw control

- ESP is built on top of ABS and TCS - it adds yaw-rate & steering sensors and control software; it can brake any one wheel even when the driver is not braking.

### Names for ESP

ESP (Bosch / Mercedes), ESC (generic), DSC (BMW, Jaguar), VSC (Toyota), VDC (Nissan, Subaru), StabiliTrak (GM).

# 11. ESP - Working & Benefits

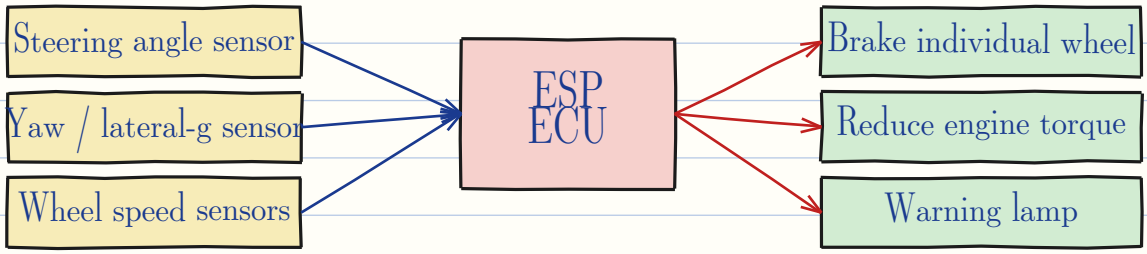



Fig 5.11 ESP block diagram

## Working

1. Driver intention (steering angle + speed) is converted to a target yaw rate.
2. ECU compares it with the measured yaw rate / lateral acceleration; difference = instability.
3. Corrective yaw moment is created by braking selected wheel(s) (up to ~ 100 bar from the pump) and reducing engine torque.
4. Once vehicle follows path, intervention stops (< 150 ms reaction). Driver feels pulsation, hears pump noise, lamp flashes.

| Situation | ESP action |
|---|---|
| Understeer (left turn) | Brake left rear wheel + cut torque |
| Oversteer (left turn) | Brake right front wheel + cut torque |
| Sudden lane change | Brake individual wheels in sequence |
| Rollover risk (SUV) | Roll-over mitigation (RMI) - brake outer front |

## Benefits

- Reduces single-vehicle crashes by ~ 30-40 %; mandatory in EU (2014), US (2012) and India for several classes; works on any road; off switch for off-road / snow chains.

# 12. Traction Control System (TCS)

- TCS (ASR, TRC, ASC) prevents the drive wheels from spinning on slippery surfaces during acceleration by limiting wheel slip -> maximum traction + stability. It is the "ABS for acceleration".

## Components

- Wheel-speed sensors (shared with ABS), ECU (integrated with ABS / ESP), hydraulic modulator with pump, electronic throttle / engine ECU interface, switch & dash lamp.

## Working

1. ECU compares driven wheel speed with non-driven wheel (vehicle) speed; slip > ~ 10-15 % = wheelspin.
2. Brake intervention: brake applied to the spinning wheel -> the differential transfers torque to the wheel with better grip (acts as an electronic limited-slip differential).
3. Engine intervention: reduce throttle opening, retard ignition, cut fuel injection to cylinders, ECU torque request.

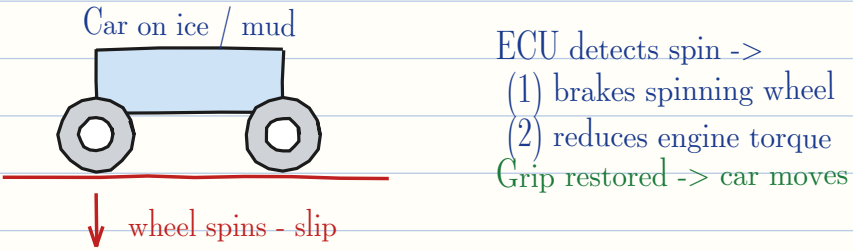



Fig 5.12 Traction control action

| System | Prevents | Operates during |
|---|---|---|
| ABS | Wheel lock | Braking |
| TCS | Wheel spin | Acceleration |
| ESP | Skid / yaw | Cornering / any |

# 13. Global Positioning System (GPS)

- GPS is a satellite-based radio-navigation system (USA, NAVSTAR) giving 3-D position, velocity and time anywhere on Earth, in all weather. Other GNSS: GLONASS, Galileo, BeiDou, IRNSS / NavIC (India).

## Segments

- Space segment: 24+ satellites in 6 orbital planes, altitude 20,200 km, period ~ 12 h. Control segment: ground stations (master control Colorado) update clocks & orbits. User segment: receivers (car, phone).

## Principle - trilateration

- Each satellite transmits a signal with its position and exact time (atomic clock). Receiver measures travel time dt -> distance d = c x dt (c = 3 x 10^8 m/s). Distances from 3 satellites fix a point on Earth; the 4th satellite corrects the receiver clock error (x, y, z, t = 4 unknowns).

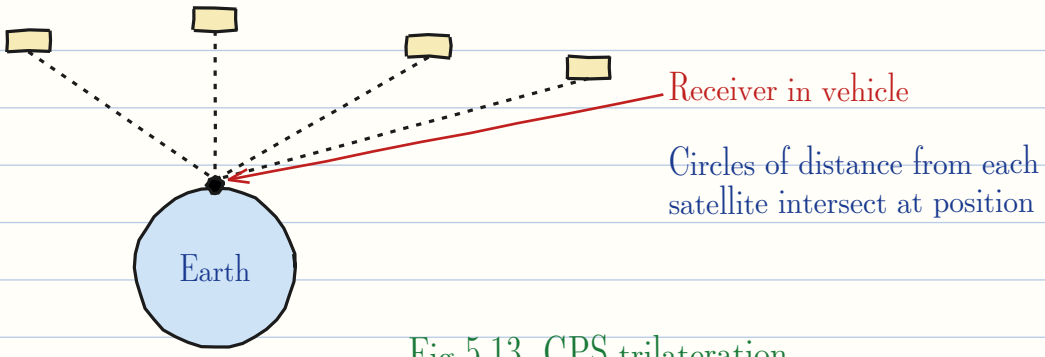



Fig 5.13 GPS trilateration

- Signals: L1 = 1575.42 MHz, L2 = 1227.6 MHz; C/A code (civil) & P code (military). Accuracy: ~ 3-5 m (civil), < 1 m with DGPS / RTK.

## Key facts

- Minimum 4 satellites needed for a 3-D fix; GPS time is kept by atomic (Cs / Rb) clocks; relativity corrections are built into the signal.
- Position fix time: cold start 30-60 s, hot start 1-5 s; update rate 1-10 Hz in car units.

**Remember**

d = c x dt; 3 satellites -> position, 4th satellite -> receiver clock correction; accuracy ~ 5 m (civil).

# 14. GPS in Automobiles

## Vehicle navigation system

- Components: GPS receiver + antenna, map database (digital map, SD / cloud), dead-reckoning sensors (gyro, odometer / wheel speed) for tunnels, processor + display + speaker, user interface (voice), map-matching software, traffic info (TMC / data).

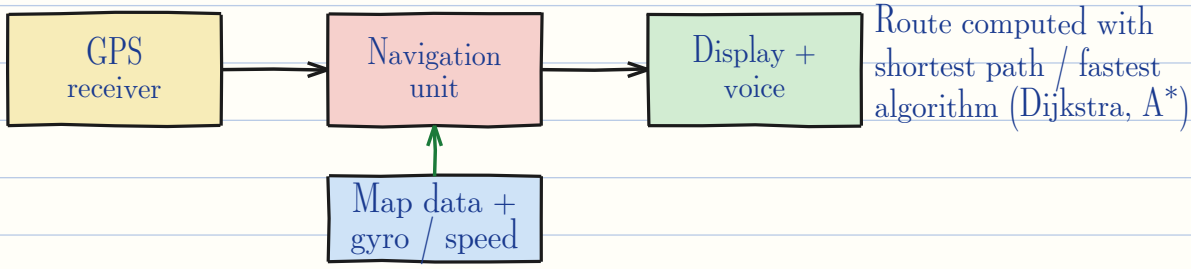



Fig 5.14 Car navigation system

## Applications in automobiles

- Turn-by-turn navigation, real-time traffic & re-routing; fleet management / vehicle tracking (AIS-140 mandatory in public vehicles in India), anti-theft recovery, emergency call (eCall / SOS), geo-fencing, speed limiting, ride-hailing & logistics (Ola, Uber), insurance telematics (pay-as-you-drive), stolen-vehicle immobilisation, ADAS & autonomous driving (with lidar, HD maps).

## Advantages / limitations

- + Saves time & fuel, improves safety & security, data for road planning. - Signal loss in tunnels / urban canyons, privacy concerns, multi-path errors, jamming / spoofing.
- Error sources: ionospheric delay, clock error, multipath, satellite geometry (DOP). Improved by WAAS / EGNOS / GAGAN (India) augmentation.

### NavIC (IRNSS)

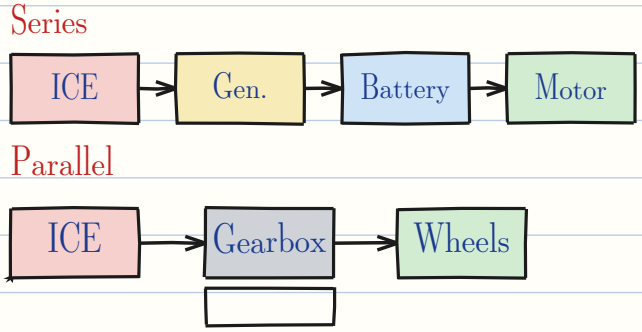
India's regional system: 7 satellites, accuracy ~ 5 m over India + 1500 km around; being fitted in vehicle tracking devices & phones.

# 15. Hybrid Electric Vehicle (HEV)

- A hybrid vehicle combines an internal combustion engine (ICE) with an electric motor / generator + battery to reduce fuel consumption & emissions while keeping range & performance.

## Classification by architecture

- Series hybrid: ICE drives only a generator; electric motor alone drives the wheels (range-extender, BMW i3 REx, Nissan e-Power). Engine runs at best-efficiency speed.
- Parallel hybrid: both ICE & motor can drive the wheels through a common transmission (Honda IMA, mild hybrids).
- Series-parallel (power-split): planetary gear splits power; best of both (Toyota Prius, Hyryder, Hycross).




Series: engine -> electricity -> motor drives wheels

Parallel: engine and motor both push the same shaft

Power-split: Prius planetary gear

Fig 5.15 Hybrid configurations

## Classification by degree of hybridisation

| Type | Motor | Function | Example |
|---|---|---|---|
| Micro | 2-5 kW | Start-stop only | Maruti SHVS |
| Mild (MHEV) | 10-20 kW | Assist + regen, no EV mode | Brezza, Ciaz SHVS |
| Full (HEV) | 30-80 kW | Electric-only driving short | Prius, Camry, City e:HEV |
| Plug-in (PHEV) | 50-100 kW | 30-80 km EV range; grid charge | Prius Prime, BYD |

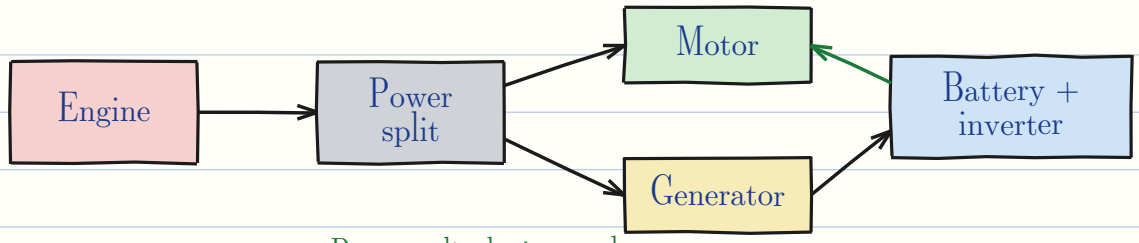
# 16. Hybrid - Components & Working Modes

## Main components

- ICE (Atkinson cycle, high efficiency), traction motor / MG (permanent magnet AC), generator, power control unit (inverter, DC-DC converter), HV battery (Ni-MH / Li-ion 200-300 V), power-split device / CVT, hybrid ECU, regenerative braking.

## Operating modes (Toyota Prius type)

| Mode | Power flow |
|---|---|
| Starting / low speed | Motor only (EV), engine OFF |
| Normal driving | Engine + motor share, engine at efficient point |
| Full throttle | Engine + motor + battery boost |
| Deceleration / braking | Regenerative braking: motor works as generator -> charges battery |
| Stop | Engine OFF (idle stop) |
| Battery low | Engine drives generator -> charges |




Power-split device = planetary gear set

Fig 5.16 Power-split hybrid drive

## Advantages

- 20-40 % better fuel economy, lower CO2 / NOx, regenerative braking (recovers 15-25 % of energy), quiet city driving, no range anxiety, no external charging (HEV).

## Disadvantages

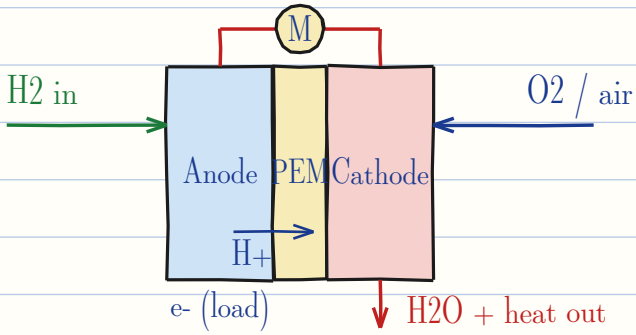
- Higher cost, extra weight, battery life / disposal, complex service & high-voltage safety (orange cables).

# 17. Fuel Cell Vehicle (FCEV)

- A fuel cell is an electrochemical cell that converts the chemical energy of hydrogen + oxygen directly into electricity (water + heat only) as long as fuel is supplied. Efficiency 40-60 % (vs ICE 25-35 %).

## PEM fuel cell (most used in cars)

- Anode (-): H2 -> 2H+ + 2e- (platinum catalyst). Cathode (+): 1/2 O2 + 2H+ + 2e- -> H2O. Overall: H2 + 1/2 O2 -> H2O + electricity + heat. Protons pass through the Proton Exchange Membrane (Nafion); electrons flow in the external circuit -> motor.




Stack of cells in series gives 200-300 V; each cell ~ 0.7 V

H2 stored at 700 bar in carbon-fibre tank (5 kg, 500 km)

Fig 5.17 PEM fuel cell

| Type | Electrolyte | Temp | Use |
|---|---|---|---|
| PEMFC | Polymer membrane | 60-80 C | Cars, buses (Mirai, Nexo) |
| AFC | KOH | 90-100 C | Space |
| PAFC | Phosphoric acid | 200 C | Stationary |
| SOFC | Ceramic | 800-1000 C | Power plants, APU |

- Advantages: zero tailpipe emission, quiet, quick refuelling (3-5 min), long range. Challenges: H2 production cost / storage, platinum cost, infrastructure, durability.